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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Hairloss
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 Decent work & growth SDG 9 Innovation SDG 12 Less waste (paperless)
Project / Solution Name	SmartDesk
Name (Leader)	Judah Teoh Em Yu
Email & Phone (Leader)	teohjudah@gmail.com | 0187765060
University Affiliation(s)	Universiti Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium-sized enterprises (SMEs), particularly university canteen vendors and small retailers, may struggle to manage large volumes of customer orders received through WhatsApp. Many businesses rely on manually reviewing messages and recording orders on paper, increasing administrative workload and the risk of overlooking important order details. This can lead to delayed responses, customer dissatisfaction, and potential lost sales.

The primary groups affected are SME owners and employees, who must spend additional time managing repetitive tasks instead of focusing on serving customers and growing their businesses. Customers are also affected when their orders are missed or responses are delayed. For small businesses with limited staff and budgets, these inefficiencies may increase labour costs and reduce operational productivity.

SmartDesk aims to address this challenge by using artificial intelligence (AI) to analyse incoming WhatsApp messages, extract relevant order information, and assist with order creation, invoice generation, payment-related processes, and automated customer responses using retrieval-augmented generation (RAG). Unlike a conventional chatbot that mainly responds to enquiries, SmartDesk focuses on transforming customer messages into actionable business information.

By reducing manual processing, SmartDesk aims to save time, lower operating costs, and improve order management efficiency. Our initial target is to reduce order-processing time by 50%, which will need to be validated through testing. Ultimately, SmartDesk seeks to make order management more efficient and affordable for SMEs, helping them serve customers better while managing their limited resources.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SmartDesk is an AI-powered application designed to help small and medium-sized enterprises (SMEs) manage customer orders received through WhatsApp more efficiently. It uses Generative AI to analyse incoming messages, identify customer requests, and extract essential order details. The system then converts this information into structured orders, assists with invoice generation and payment-related processes, and provides automated responses using Retrieval-Augmented Generation (RAG).

Unlike conventional chatbots that primarily answer customer enquiries, SmartDesk focuses on automating the order management workflow, reducing repetitive manual tasks and minimising the risk of overlooked orders. Unclear information can be flagged for human verification to improve accuracy and reliability.

To develop SmartDesk, our team intends to leverage technical skills gained from the SDC Phase 1 workshops, including Generative AI APIs for message understanding and information extraction, database architecture for storing and managing customer and order records, and cloud hosting for reliable access to the application. We also aim to integrate WhatsApp messaging and payment-related services to support a more seamless ordering process.

Ultimately, SmartDesk aims to simplify daily business operations, reduce administrative workload, and help SMEs manage customer orders more efficiently and affordably.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Orbyte SmartDesk is built on Node.js using only built-in modules, with a vanilla HTML, CSS and JavaScript owner dashboard and a SQLite database storing customers, messages, orders, invoices, settings and a full audit log. Messages arrive through the WhatsApp Business Cloud API webhook, and approved invoices sync to Orbyte through a REST connector, currently mocked until Orbyte's API is available. The pipeline is simple: a WhatsApp message triggers the flow, data is saved, and an LLM classifies intent and extracts order details. Routine replies use the owner's own settings, while other questions pass through our RAG pipeline: FAQ knowledge base, meaning-based and keyword search, an AI-drafted answer, and code checks on prices and sources. The owner approves drafts and invoices, replies and follow-ups return via WhatsApp, and every action is logged, with fallbacks keeping the shop running if the AI fails. Our GenAI engine is the Google Gemini API (gemini-3.8-flash and gemini-embedding-001), used with our own key. Yes, we used the SDC sponsor-provided cloud platform, Replit, to develop and deploy the prototype. Replit gave us a ready cloud environment with no local setup: we imported the code from GitHub, Replit Agent configured the run environment, and API keys were kept safe in Replit Secrets. Its public web address replaced temporary tunnels as our WhatsApp webhook URL, Replit's invite feature lets teammates work in the same project, and one-click publishing keeps a live demo available for testing and judging.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied the five design-thinking stages, and each one shaped the product. To empathise, we interviewed an SME canteen employers with couple of questions, learning that orders arrive as messy multilingual WhatsApp chats, sometimes invoices are retyped, payments are chased by hand, and owners refuse bots that message customers without approval. We then defined one problem: WhatsApp messages that never become correct invoices and paid sales, dropping e-invoicing after finding most small SMEs are exempt. During ideation we rejected POS ordering and a general chatbot, choosing a design where AI reads and drafts, the owner's data decides prices, and the owner approves what matters. We prototyped quickly, from OrderLah to SmartDesk, using a Customer Simulator to test safely. Testing drove real fixes: wrong-language replies, missed second questions and slow AI led to language detection, question splitting, time limits and fallbacks, now guarded by 42 automated tests. The result is technically viable (no third-party dependencies, deployed on Replit, works even if AI fails) and user-friendly (customers simply use WhatsApp; owners approve with one tap). For Phase 2, we will pilot with real SMEs, connect live WhatsApp and Orbyte APIs, and add owner login and webhook security.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.2 (productivity through technological upgrading and innovation) and Target 8.3 (growth of micro-, small- and medium-sized enterprises). SmartDesk automates WhatsApp sales admin for micro F&B sellers like Mak Teh Kek Lapis.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Every enquiry, invoice and payment reminder is handled manually: 0% automated replies, 0% systematic follow-ups, and 36% of a 45.5-hour week spent on admin (about 16.4 hours). Enquiries average 42 hours for a reply.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% of enquiries answered within 1 minute (tracked in SmartDesk's dashboard and audit log); weekly owner admin hours.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	At least 80% of routine enquiries answered within 1 minute; 100% of quiet customers followed up; 30% fewer admin hours.

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes /☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes /☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes /☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] JUDAH

[Date] 24/10/2026